

$$\begin{aligned}
 b) \quad s &= \frac{N_s - N_r}{N_s} \\
 &= \frac{1500 - 1440}{1500} \\
 &= \frac{60}{1500} \\
 &= 0.04 = 4\%
 \end{aligned}$$

$$\begin{aligned}
 c) \quad f_r &= s \cdot f \\
 &= 0.04 \times 50 \\
 &= 2 \text{ Hz}
 \end{aligned}$$

4.) A magnetic circuit is similar to an electric circuit because both involve a flow driven by a force and opposed by a resistance-like parameter.

Electric Circuit	Magnetic Circuit
• Current (I)	Magnetic flux (ϕ)
• Voltage (V)	MMF (NI)
• Resistance (R)	Reluctance (R)
• Conductance	Permeance
• Ohm's law: $V = IR$	$NI = \phi R$

Similarities:-

i) Driving Force:-
 Electric: Voltage
 Magnetic: MMF

ii) Flow Quantity
 Electric: Current
 Magnetic: Flux

iii) Opposition:-
 Electric: Resistance